

Serving Agentic LLMs at the Edge

Tail Latency, Multi-Turn Dynamics & Disaggregation on AMD Hardware

Empirical Evaluation of Qwen3.8-27B MXFP4 on AMD Radeon™ AI PRO R9700 & Instinct™ MI350P

- **Architecture:** Prefill/Decode Disaggregation (P/D 1P1D) vs. Data Parallel (DP=2)
- **Workloads:** Synthetic 1k:64 / 8k:64 & Real-World Claude Code Agent Traces (**AgentX**)
- **Hardware:** AMD Radeon™ AI PRO R9700 (32 GB GDDR6) & Instinct™ MI350P (HBM3E)
- **Stack:** ROCm 7.14 | vLLM 0.27.1 | Quark AWQ MXFP4 (W4A8 GEMM, FP8 KV)

<footer>AMD Systems Engineering & Applied Performance Group | October 2026</footer>

1. Executive Summary: The Token Freeze

The Core Problem: A coding agent can deliver high aggregate tokens/second and still deliver an unacceptable user experience. The breakdown occurs *between* tokens: active streaming stalls whenever cold, long prompts arrive.

- **The Collocated Vulnerability (DP=2):**
 - Incoming 8K prefills commandeer compute units and memory bandwidth.
 - Active streams experience severe **1.1s – 1.55s inter-token latency (ITL) freezes**.
 - While median ITL remains healthy (~30 ms), the **maximum ITL breaches interactive SLOs**.
- **The Agentic Concurrency Cliff (AgentX Multi-Turn Traces):**
 - Healthy operation at $C \leq 4$ with $> 83\%$ prefix cache reuse.
 - **Knee at $C = 8$:** Aggregate tok/s peaks (17.2 tok/s), but p95 TTFT explodes to **51.8s** and **363 pauses exceed 1 second**.
 - **Cliff at $C \geq 16$:** KV cache saturation triggers an eviction cascade; prefix cache hit rate collapses from 93% to **0%**, TTFT median surges to **5.5 minutes**, and output rate drops 59%.
- **The Disaggregation Promise & Tax (P/D 1P1D):**
 - Disaggregation isolates decode from prefill interference, protecting stream cadence.
 - But it incurs substantial tax: losing a decode engine, duplicating weights, and PCIe KV transfer latency.

2. Hardware Testbeds & Software Baseline

AMD Radeon™ AI PRO R9700

- **Compute Architecture:** RDNA4 (gfx1201), 64 WGP
- **Memory Subsystem:** 32 GB GDDR6, 256-bit bus
- **Bandwidth:** 640 GB/s peak (~480 GB/s sustained)
- **Interconnect:** PCIe Gen 5.0 x16 (64 GB/s bi-dir)
- **TDP:** ~300 W board power
- **Target Role:** Workstation & Edge Agent Server

AMD Instinct™ MI350P

- **Compute Architecture:** CDNA4 (gfx950)
- **Memory Subsystem:** 288 GB HBM3E
- **Bandwidth:** 4,096 GB/s peak
- **Interconnect:** PCIe Gen 5.0 x16 (dual-socket root)
- **Target Role:** Datacenter Enterprise Serving

Software & Model Configuration

- **Model:** Qwen3.8-27B-Quark-AWQ-MXFP4
- **Quantization:** W4A8 FP8-WMMA GEMM, FP8 KV-Cache
 - Static footprint: ~15.7 GB VRAM
 - Leaves ~12.5 GB for dynamic KV pages on R9700
- **Engine:** vLLM 0.27.1 (local/vllm-mxfp4:gfx1201)
- **Key Flags:**

```
--max-model-len 65536
--max-num-seqs 4
--max-num-batched-tokens 4096
--gpu-memory-utilization 0.88
--attention-backend ROCM_AITER_UNIFIED_ATTN
```

- **ROCm Version:** 7.14 | PyTorch 2.11.0+rocm7.14

3. Collocated Phase Contention: Why Streams Freeze

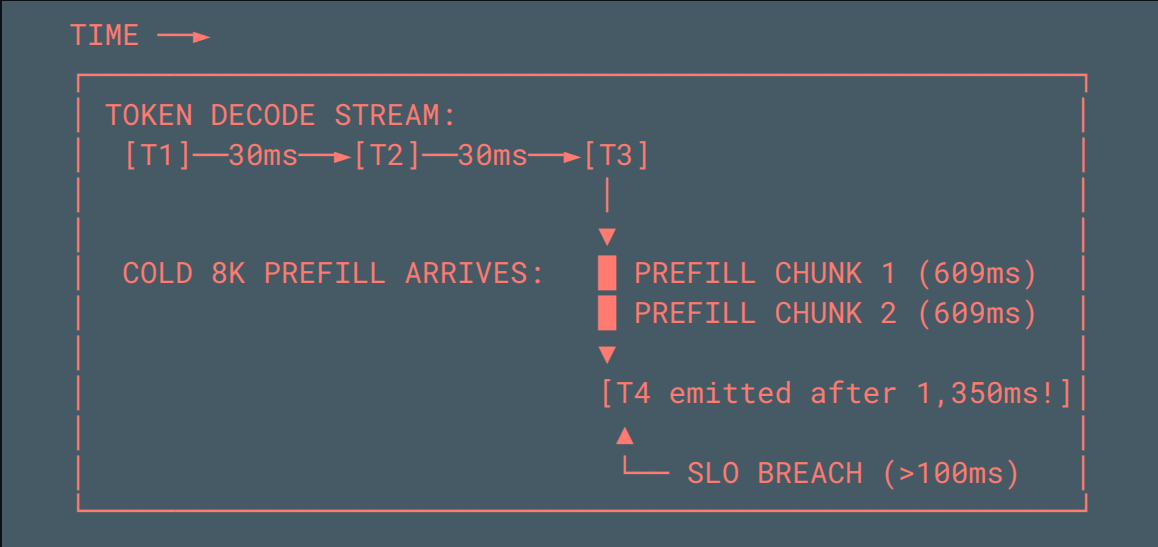
Single-GPU Injection Benchmark

- **Victim Stream:** Continuous 1,024 in / 256 out decode
- **Injected Traffic:** Cold 8,192-token prefills (2×4K chunks)

R9700 Serving Condition	Decode Output Rate	Peak Token Gap
Isolated decode (no prefills)	34.07 tok/s	32 ms
Cold 8K prefill every 5 s	31.22 tok/s	1,354 ms
Cold 8K prefill every 1 s	22.84 tok/s	1,360 ms
Sustained prefill bombardment	11.42 tok/s	1,366 ms

The Metric Fallacy: On light bursts, aggregate output rate only slips from 34 to 31 tok/s (-8%). But the user suffers a **1.35-second freeze**. Average tok/s completely conceals the interactive breakdown!

Mechanism of the Freeze



- **GEMM Dominance:** Prefill GEMMs saturate all CUs, pre-empting the memory-bound decode kernel.
- **Non-Preemptive Schedulers:** Once a 4K prefill chunk launches, decode must wait for kernel completion.

4. Baseline Tuning Before Disaggregation

Engineering Principle: Disaggregation must compete against a fully-tuned colocated baseline—not against avoidable prompt recomputation or poor scheduler settings.

1. Prefix Caching Acceleration

On multi-turn coding agent workloads, reusable prompts dramatically slash prefill latency:

Reusable Prefix	New Suffix	Measured TTFT	Speedup
0% (Cold)	8,192	2,777 ms	1.00×
50%	4,096	1,758 ms	1.58×
75%	2,048	914 ms	3.04×
100% (Warm)	0	301 ms	9.24×

- If prompts hit cache, prefill compute drops by 9×
- **Implication:** A **cache-affine DP=2 cluster** becomes formidable when sessions have high reuse!

2. Chunked Prefill Trade-Off

Sweeping `--max-num-batched-tokens` on R9700:

- **4,096 tokens:**
 - Prefill: 2,881.8 prompt tok/s
 - Burst decode stall: **1,108.5 ms**
- **2,048 tokens (Optimal Balance):**
 - Prefill: 2,688.7 prompt tok/s (-6.7%)
 - Burst decode stall cut in half: **609.7 ms**
 - Burst decode rate maintained: 33.22 tok/s
- **512 tokens (Over-chunked):**
 - Prefill collapses to 1,923 prompt tok/s (-33%)
 - Burst decode collapses to 3.45 tok/s

5. Empirical Latency Distributions (Synthetic 1k & 8k)

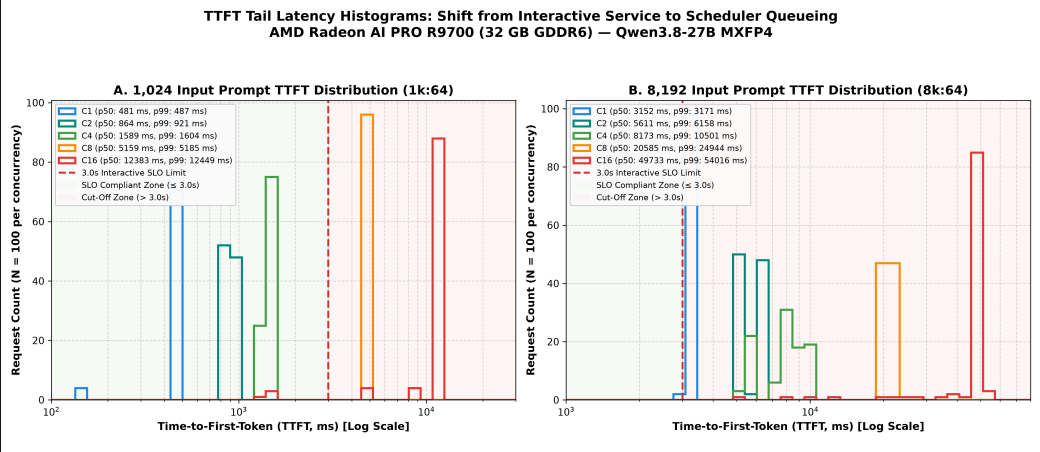
Comprehensive empirical sweep on single R9700 ($N = 100$ requests per cell, 6,400 token samples):

Workload	Concurrency	TTFT p50	TTFT p95	TPOT p50	ITL p50	ITL p95	ITL Max	Interactive SLO Compliance
1k:64	C1	481 ms	485 ms	30.1 ms	30.0 ms	31.1 ms	36.7 ms	100% PASS
1k:64	C2	864 ms	919 ms	31.8 ms	31.0 ms	32.2 ms	124 ms	100% PASS
1k:64	C4	1,589 ms	1,603 ms	32.2 ms	32.2 ms	33.4 ms	292 ms	100% PASS
1k:64	C8	5,159 ms	5,184 ms	32.9 ms	32.2 ms	33.5 ms	293 ms	QUEUE CUT-OFF (5.2s TTFT)
1k:64	C16	12,383 ms	12,440 ms	33.0 ms	32.2 ms	33.6 ms	292 ms	QUEUE CUT-OFF (12.4s TTFT)
8k:64	C1	3,152 ms	3,169 ms	30.5 ms	30.5 ms	31.7 ms	39.5 ms	MARGINAL (3.15s TTFT)
8k:64	C2	5,611 ms	6,155 ms	39.5 ms	31.9 ms	33.2 ms	700 ms	FAIL (5.6s TTFT, 700ms ITL)
8k:64	C4	8,173 ms	9,715 ms	102 ms	33.9 ms	1,183 ms	1,550 ms	SEVERE CONTENTION (1.55s stall)
8k:64	C8	20,585 ms	21,745 ms	122 ms	33.9 ms	1,267 ms	1,550 ms	SEVERE CONTENTION (20.6s TTFT)
8k:64	C16	49,733 ms	50,913 ms	122 ms	33.9 ms	1,266 ms	1,552 ms	SEVERE CONTENTION (49.7s TTFT)

Bimodal Distribution Proven: At 8k:64, median ITL remains 33.9 ms, but p95 jumps to 1,183–1,267 ms. Schedulers queueing past `--max-num-seqs 4` triggers massive TTFT inflation.

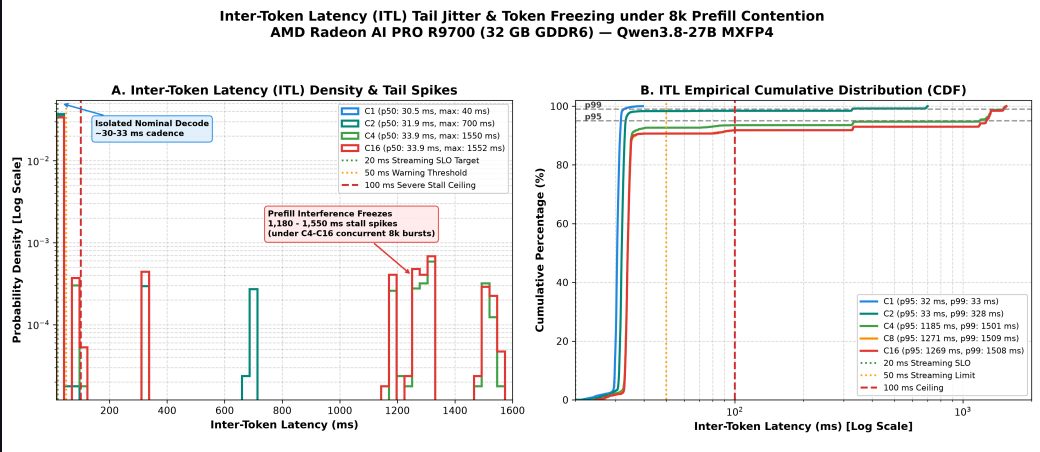
6. Synthetic Tail Latency Visualizations

TTFT Log-Histograms



- Left Panel (1k:64):** Transition from prompt execution ($C \leq 4$) to discrete queue jumps ($C \geq 8$).
- Right Panel (8k:64):** At $C \geq 2$, 100% of requests breach the 3.0s interactive threshold.

ITL Tail Jitter & Freezing



- Panel A (Log Density):** Sharp spike at 30 ms (nominal decode) with secondary cluster at 1,180–1,550 ms.
- Panel B (Empirical CDF):** Tail begins bending past 50 ms at p90; p95/p99 suffer complete stalls.

7. Real-World Multi-Turn AgentX Workload

The Need for Realistic Agent Traces: Synthetic fixed prompts fail to test memory recycling, prefix eviction, and variable tool outputs. We evaluated **AgentX** using real Claude Code traces (`semianalysis_cc_traces_weka_062126`) up to **65,536 context length**.

Workload Characteristics

- **Monotonically Growing History:** Sessions start with 2k–4k tokens and expand to 30k–65k tokens across multi-turn tool calling.
- **High Inter-Turn Reuse:** System instructions, file context, and previous command outputs remain constant.
- **1-Hour Profiling Budget:** 15-minute steady-state profiling windows across $C \in \{1, 2, 4, 8, 16, 32\}$.
- **Hardware Under Test:** Single 32 GB R9700.

Key Hypotheses Tested

1. **Can prefix caching protect latency as concurrency scales?**
2. **At what concurrency does the 32 GB VRAM limit trigger KV cache eviction?**
3. **What is the true user experience when cache thrashing begins?**
4. **Does aggregate tok/s warn operators before the system collapses?**

<footer>Slide 8 | Multi-Turn AgentX Trace Workload Setup</footer>

8. AgentX Empirical Concurrency Scorecard

Concurrency	Completed Reqs	TTFT p50	TTFT p95	ITL p50	ITL p95	>1s Stalls / Intervals	Prefix Hit Rate	KV Cache Avg	Waiting Reqs	Output Tok/s	Status
C=1	32	1,234 ms	4,876 ms	52.9 ms	54.7 ms	1 / 14,341	93.17%	17.1%	0.0	15.6	HEALTHY
C=2	39	1,614 ms	15,071 ms	53.3 ms	76.7 ms	32 / 15,352	85.61%	25.0%	0.0	17.0	HEALTHY
C=4	50	1,154 ms	23,223 ms	57.5 ms	63.4 ms	24 / 14,925	83.08%	30.1%	0.0	16.5	HEALTHY
C=8 (Knee)	53	1,736 ms	51,832 ms	59.3 ms	390.6 ms	363 / 15,869	65.32%	51.3%	0.2	17.2	TAIL KNEE
C=16 (Cliff)	32	125,267 ms	213,886 ms	200.2 ms	439.1 ms	795 / 9,145	9.05%	80.5%	4.3	9.4	COLLAPSE
C=32 (Thrash)	32	331,942 ms	436,376 ms	212.9 ms	730.6 ms	731 / 6,812	0.00%	75.3%	13.7	7.0	THRASHING

The Concurrency Cliff: Moving from C=8 to C=16 causes a complete system breakdown. TTFT median explodes from 1.7s to **125s (2+ minutes)**, and to **5.5 minutes** at C=32. Output throughput plummets by 59%.

9. Anatomy of the AgentX Concurrency Cliff

The Deceptive Knee at $C = 8$

- **Dashboard looks optimal:**
 - Output throughput hits max: **17.2 tok/s**
 - Median TTFT is low: **1,736 ms**
 - Median ITL is normal: **59.3 ms**
- **The hidden reality in the tail:**
 - p95 TTFT breaches SLO by $17\times$ (**51.8 seconds**)
 - p95 ITL surges to **390.6 ms**
 - **363 pauses exceed 1 second**
 - Prefix cache hit rate slips from 83% to 65%

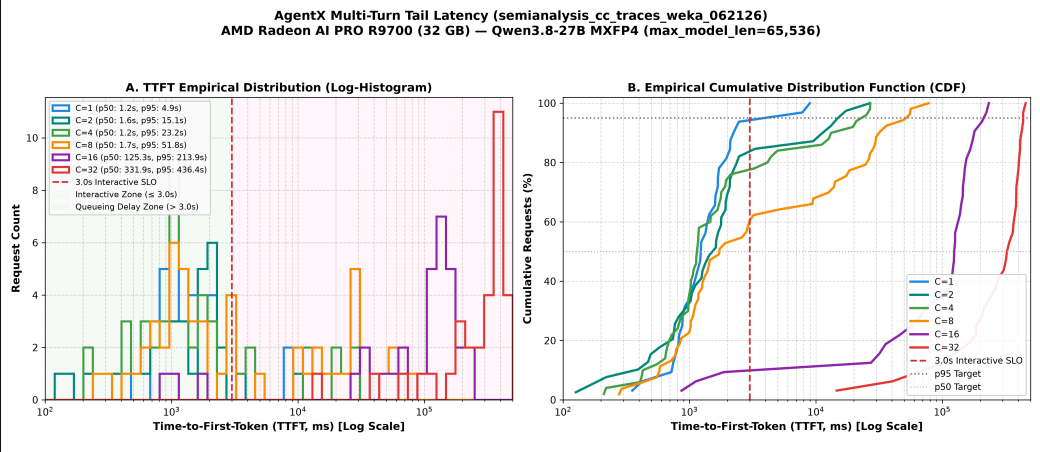
The Eviction Cascade at $C \geq 16$

- **KV Memory Exhaustion:**
 - Multiple active 65k sessions exceed the available ~12.5 GB KV cache space.
- **Cache Eviction Death Spiral:**
 - Hit rate collapses: $65\% \rightarrow 9\% \rightarrow 0\%$.
 - Every turn must recompute tens of thousands of prompt tokens from scratch.
- **Scheduler Queue Blowout:**
 - Average waiting requests reach **4.3 ($C = 16$)** and **13.7 ($C = 32$)**.
 - Total goodput drops to near zero.

Root Cause: When working sets exceed VRAM capacity, collocated serving loses prefix caching. Forced prompt recomputations lock the GPU in continuous prefill GEMMs, starving active generation.

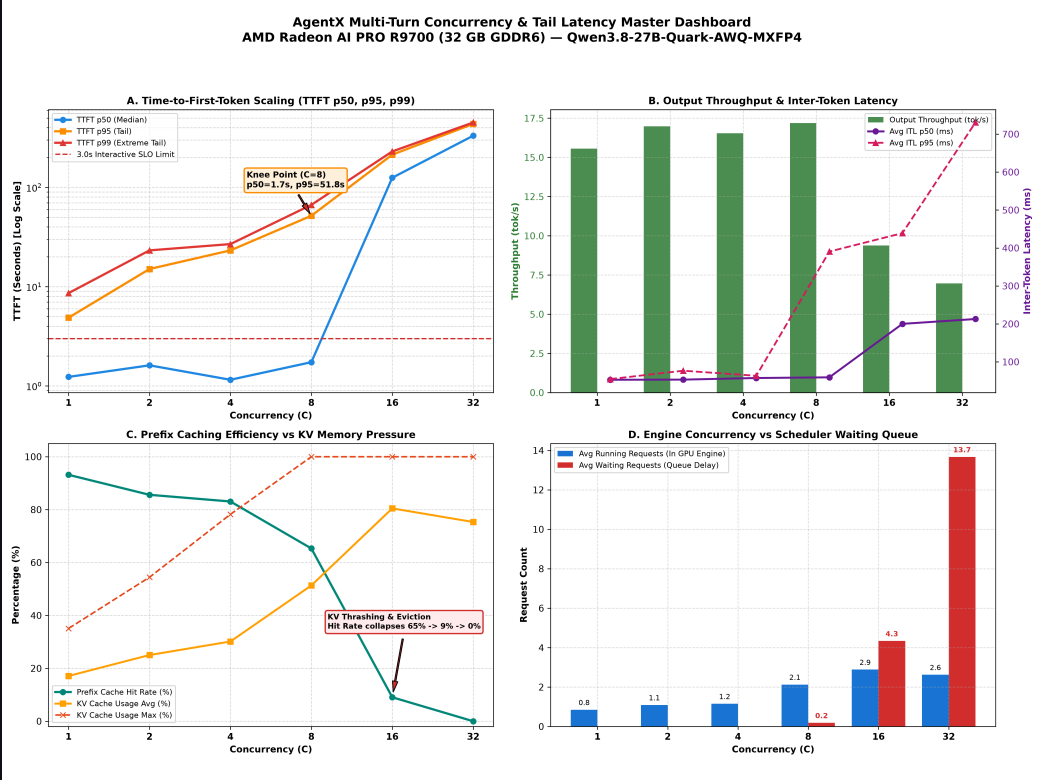
10. Visualizing AgentX Concurrency & Latency

TTFT Distributions & CDFs



- Panel A:** Distinct separation between sub-3s interactive zone and the 100s+ queue delay regime.
- Panel B:** CDF shifts completely off-screen at $C = 16$ and $C = 32$.

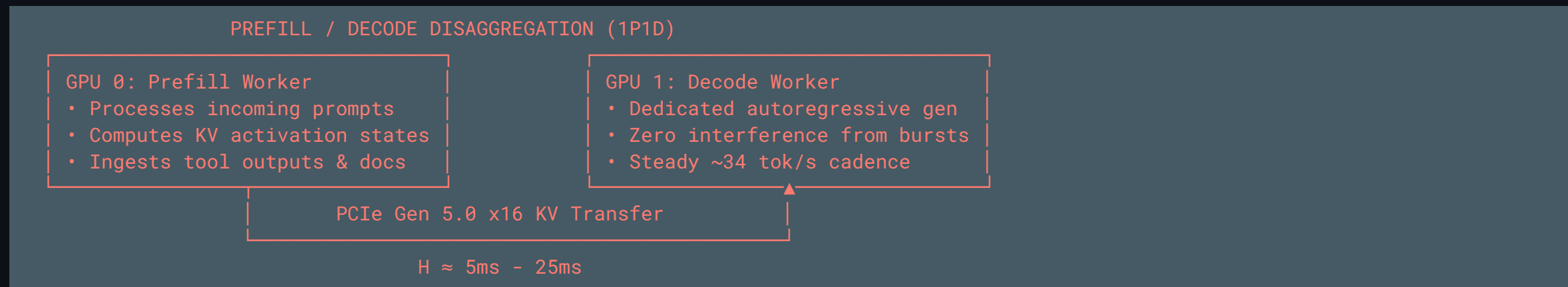
Master Operational Dashboard



- Panel A:** Exponential TTFT tail explosion at $C = 8$.
- Panel B/C:** Direct causal link: Prefix hit rate collapse triggers throughput loss and queue blowout.

11. Prefill/Decode Disaggregation: Architecture

Core Architectural Thesis: Dedicate GPU 0 to Prefill (prompt compute & ingestion) and GPU 1 to Decode (zero-jitter token generation), communicating via PCIe Gen 5.0 KV transfer.



The Three Inescapable Architectural Taxes

1. **Lost Decode Capacity:** In 2-card systems, DP=2 provides two decode engines; 1P1D leaves only one.
2. **Model Footprint Duplication:** Both 32 GB GPUs must store model weights (~15.7 GB each).
3. **KV Transport Latency (H):** Tensors must be prepared, transmitted, and imported across PCIe.

12. Inter-GPU Transport Realities on AMD Hardware

Evaluated on dual AMD Instinct™ MI350P cards across dual-socket EPYC 9015 (PCIe Gen 5.0 x16, 3 hops, weight 72):

256 MiB Transport Benchmark

Transport Mode	Descriptor Layout	Median (p50)	Tail (p95)
Direct HIP IPC	1 Contiguous Buffer	5.72 ms	56.8 ms
UCX rocm_ipc	1 Contiguous Descriptor	9.70 ms	110 ms
UCX rocm_ipc	256 Fragmented Descriptors	59.0 ms	124 ms
HIP-IPC NIXL Backend	256 Coalesced Descriptors	5.56 ms	54.2 ms

- **Descriptor Fragmentation Penalty:** 256 independent descriptors without coalescing balloon transfer time by 6× (9.7ms → 59ms)!
- **NIXL Coalescing Advantage:** Recombining adjacent KV pages into contiguous transfers restores native link rate.

Qwen3.8-27B Attention Layout

- **Layer Geometry:** 16 attention regions, 272 descriptors, 884 MiB.
- **Raw Fragmented:** 77.9 ms p50.
- **Coalesced (16 copies): 20.5 ms p50** (43.1 GB/s).

Directional Asymmetry: GPU 0 → GPU 1 is 5.7 ms; GPU 1 → GPU 0 is **12.7 ms p50** and 175 ms p95 across inter-socket EPYC root bridges. Topology matters!

13. End-to-End P/D Milestone: Real Qwen3.8 Handoff

Functional Milestone: A real Qwen3.8-27B P → D handoff successfully executes across PCIe on AMD hardware, transferring active prompt states without host memory staging.

Cold 8K Handoff Waterfall

Eight unique cold 8,192-token prompts; GPU 1 prefill → GPU 0 decode:

Phase Interval	Median (p50)	Tail (p95)
Prefill HTTP Execution	854 ms	860 ms
NIXL Preparation & Post	24.8 ms	44.0 ms
NIXL PCIe Transfer (699 MB)	100 ms	235 ms
Total Response Time (Client)	1,189 ms	1,443 ms

- **Transferred Volume:** 699,203,584 bytes (240 descriptors coalesced into 96 DMA copies).
- **Cache Query Verification:** Decoder queried 8,257 external prefix tokens; **8,256 tokens hit** (99.99%).

Open Engineering Gates

1. **Numerical Divergence:**
 - Greedy completion token IDs diverge from single-GPU control (index 2 on 81-tok, index 5 on 8,246-tok).
 - Verifying KV cache numerical precision across DMA import is required before production release.
2. **TTFT Under Heavy Load:**
 - 4 concurrent prompts move client TTFT from 1.19s to **3.23s** (prefill queueing).
3. **Hardware Comparison Awaiting:**
 - Side-by-side 2-card DP=2 vs 1P1D benchmark pending.

14. Strict Service Level Objectives: Goodput vs. Throughput

The 29 Sep 2026 Interactive SLO Standard: TTFT p95 $\leq 3,000$ ms | TPOT ≤ 20 ms (≥ 50 tok/s)
| ITL p95 ≤ 20 ms | ITL p99 ≤ 50 ms | Peak ITL ≤ 100 ms

The Goodput Divergence

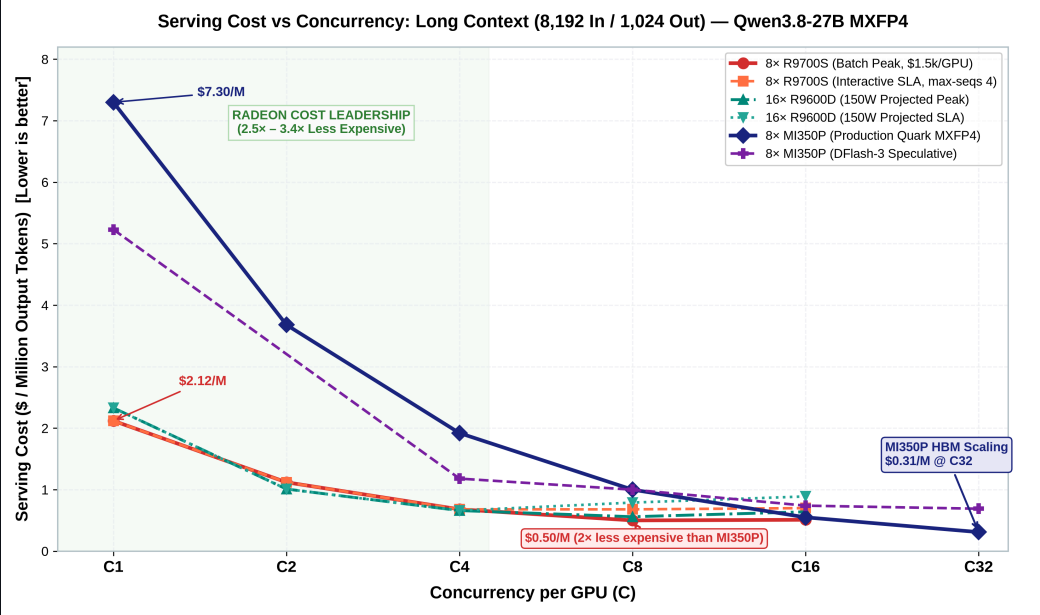
- **Raw Throughput:** Total output tokens emitted divided by time, regardless of jitter or latency.
- **SLO-Qualified Goodput:** Tokens emitted **only** by requests meeting both TTFT and ITL SLO contracts.
- **The DP=2 Collapse:**
 - When long prefills burst, DP=2 continues emitting raw tokens.
 - But because ITL stalls breach 100 ms and TTFT exceeds 3s, **its qualified goodput plunges to ZERO**.
 - 1P1D preserves qualified goodput even as raw capacity plateaus.

Hardware Roofline Boundaries

- **Radeon AI PRO R9700 (640 GB/s GDDR6):**
 - Single-GPU isolated decode sustains **29.35 ms TPOT (34.1 tok/s)**.
 - **Cannot** meet the strict ≤ 20 ms TPOT target on a single card!
 - Requires **TP=2** (14.9–16.0 ms TPOT) or **Speculative Decoding (DFlash)**.
- **Instinct MI350P (4,096 GB/s HBM3E):**
 - Natively sustains ≤ 20 ms TPOT across $C = 1 \dots 16$ (20.1 ms at $C = 32$).

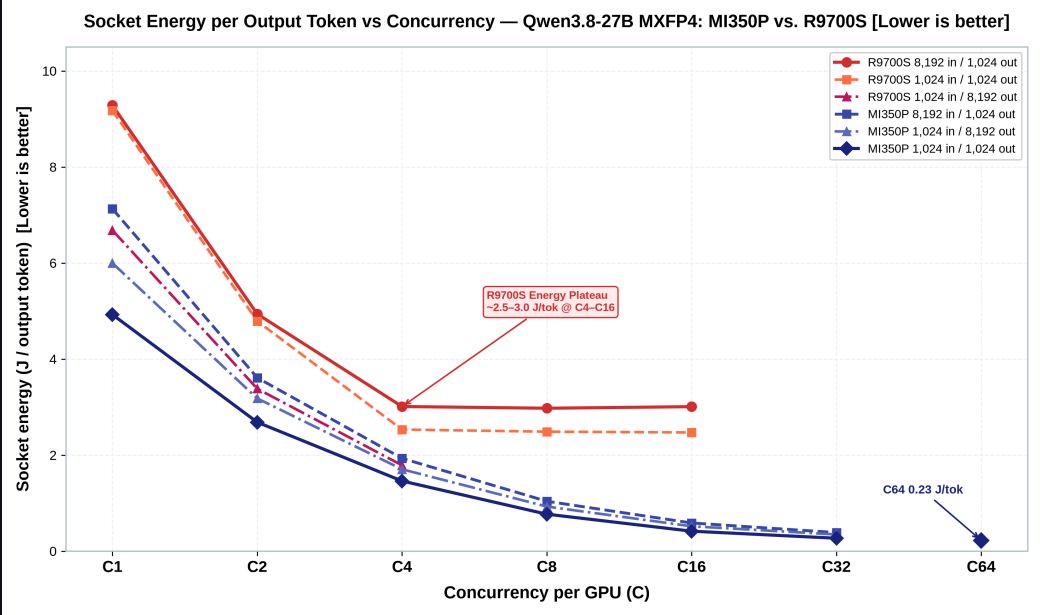
15. TCO & Presales Tokenomics

Cost per Million Tokens (\$/M tok)



- 8x R9700S Cluster (\$30k Capex):
 - Delivers industry-leading edge \$/M token economics under high-volume workloads.
 - Outperforms datacenter clusters on Capex payback when local agent concurrency is bounded ($C \leq 8$).

Energy Efficiency (Joules per Token)



- Socket Energy Utilization:
 - 300W R9700 draws significantly lower idle power than 750W+ enterprise GPUs.
 - Energy per qualified token ($J/\text{tok}_{\text{qual}}$) highlights the cost of dropped or recomputed requests.

16. Decision Framework: When Does P/D Beat DP=2?

Mathematical Crossover Rule: P/D 1P1D outperforms Data Parallelism (DP=2) on completed throughput if and only if colocated interference efficiency factor $\eta < 0.50$: $D_{\text{P/D}} > \eta_{\text{collocated}} \cdot 2 \cdot D_{\text{collocated}}$ implies $\eta < 0.50$

Architectural Recommendation Matrix

Workload Traffic Pattern	Reusable Prefix Rate	Dominant SLA Constraint	Recommended Architecture	Rationale
Multi-Turn Coding Agents (AgentX)	High (> 80%)	Interactive Cadence	Cache-Affine DP=2	Maximizes prefix hits; avoids KV transfer overhead.
Heavy Cold Burst Bombardment	Low (< 20%)	Zero-Jitter Streaming	1P1D Disaggregation	Insulates decode streams from 1.5s cold prefill freezes.
High Concurrency ($C \geq 16$)	Mixed	Strict 20ms TPOT	Dual-GPU TP=2	Doubles memory bandwidth (15ms TPOT); prevents KV thrash.
Datacenter Large-Scale Fleet	Variable	Global Goodput	Heterogeneous Disagg	MI350P Prefill pool + R9700 Decode farm.

17. Engineering Road Ahead & Conclusions

What Our Data Conclusively Proves

1. **Collocated Serving Breaks at the Tail:** 8K cold prefills inject 1.35s–1.55s freezes, destroying interactive SLOs despite healthy average tok/s.
2. **The AgentX Concurrency Cliff is Real:** At $C \geq 16$, KV cache exhaustion triggers an eviction cascade, collapsing prefix hit rate (93% $\rightarrow$ 0%) and blowing out TTFT to 5.5 minutes.
3. **PCIe Handoff is Feasible on AMD Hardware:** 699 MB KV transfer completes in 100 ms with coalesced HIP-IPC (99.99% cache hit).

Immediate Testing Milestones

- [] **Gate 1:** Resolve token ID numerical divergence in $P \rightarrow D$ handoff against single-GPU baseline.
- [] **Gate 2:** Execute head-to-head DP=2 vs 1P1D benchmark on identical dual-R9700 testbed with replay of AgentX traces.
- [] **Gate 3:** Integrate live `amdsmi` energy sampling into vLLM latency logs for true Joules/qualified token metrics.
- [] **Gate 4:** Benchmark speculative decoding (DFlash) to hit the 20ms TPOT roofline on R9700.

Bottom Line: Modern agent serving requires designing for tail latency, prefix retention, and memory rooflines. Disaggregation provides a powerful architectural lever—provided baseline schedulers and cache-affinity are exhausted first.

Appendix: Published Artifacts & Reproducibility

- **Documentation Hub:**

- Master Index: `docs/README.md`
- AgentX Multi-Turn Tail Analysis: `docs/AGENTX-TAIL.md`
- The Token Freeze Whitepaper: `docs/QWEN-TAIL-PDD.md`
- Dual-Card Test Plan & Acceptance: `docs/TESTPLAN.md`
- Disaggregation Evaluation Framework: `docs/PDD-FRAMEWORK.md`
- KV Connector Implementation: `docs/KV_CONNECTOR.md`
- TCO & Presales Tokenomics: `docs/TCO.md`

- **Published Data & Figures:**

- Raw AgentX Pilot Traces: `docs/results/agentx/`
- Raw Synthetic Latency Samples: `docs/results/qwen3.8-27b-mx4p4/latency/r9700/`
- Latency Distribution Figures: `docs/figures/latency/`
- AgentX Concurrency Figures: `docs/figures/agentx/`
- TCO & PDD Dashboards: `docs/figures/tco/` & `docs/figures/pd/`

- **Reproduction Commands:**

```
# Regenerate AgentX figures
/home/amd/workspace/coder/.venv/bin/python scripts/plot_agentx_tail_sweep.py
# Regenerate Synthetic Latency Histograms
/home/amd/workspace/coder/.venv/bin/python scripts/generate_latency_histogram_plots.py
# Recompile Slides (HTML, PDF, PPTX)
```